Ex. No. 4: OFDMA Resource Block Allocation

AIM

To simulate and analyze **OFDMA (Orthogonal Frequency Division Multiple Access)** resource block allocation and evaluate system performance in terms of **resource utilization** and **user throughput** using MATLAB.

Objective 1: Resource Block Allocation in OFDMA

Aim

To determine the number of **Resource Blocks (RBs)** available in an OFDMA system based on the system bandwidth and allocate them efficiently among multiple users.

MATLAB File Name

exp4_1_OFDMA_Resource_Block_Allocation.m

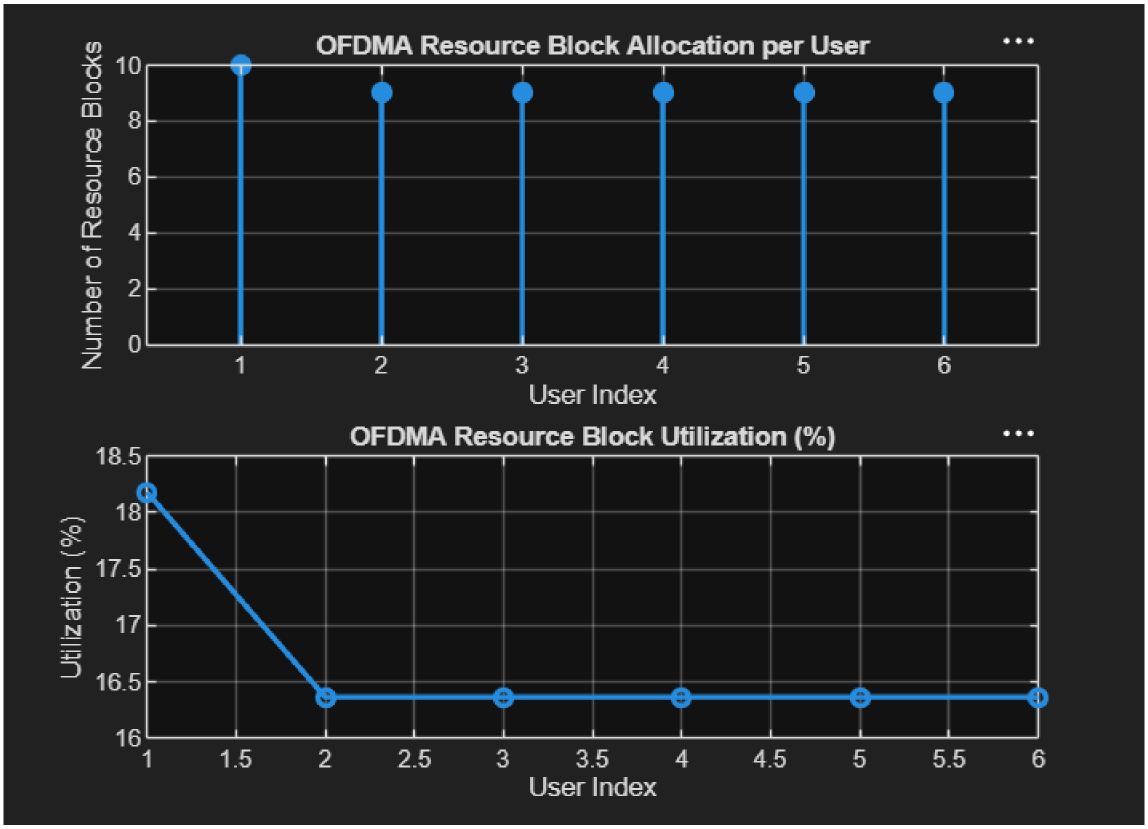
MATLAB Code

```
clc;

clear;

close all;


% System parameters

bandwidth = 10e6;      % 10 MHz

rb_size = 180e3;       % 180 kHz per RB


% Total Resource Blocks

total_RB = floor(bandwidth/rb_size);


% Users

users = 1:6;


% RB allocation
```

```
base = floor(total_RB/length(users));
alloc = base*ones(1,length(users));

alloc(1:mod(total_RB,length(users))) = ...
    alloc(1:mod(total_RB,length(users))) + 1;

% Utilization
utilization = (alloc/total_RB)*100;

disp('Total Resource Blocks:')
disp(total_RB)

disp('RB Allocation per User:')
disp(alloc)

figure

% Graph 1
subplot(2,1,1)
stem(users,alloc,'filled','LineWidth',2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on

% Graph 2
subplot(2,1,2)
```

```
plot(users,utilization,'-o','LineWidth',2)

title('OFDMA Resource Block Utilization (%)')

xlabel('User Index')

ylabel('Utilization (%)')

grid on
```

Output



- Total available Resource Blocks.
- Resource Block allocation for each user.
- Graph of RB Allocation per User.
- Graph of RB Utilization (%).

Result

The available resource blocks were successfully distributed among multiple users. The utilization graph shows efficient spectrum sharing with balanced allocation across users.

Objective 2: Resource Block Utilization Analysis

Aim

To evaluate the **Resource Block Utilization (%)** by calculating the percentage of allocated RBs for each user in an OFDMA system.

MATLAB File Name

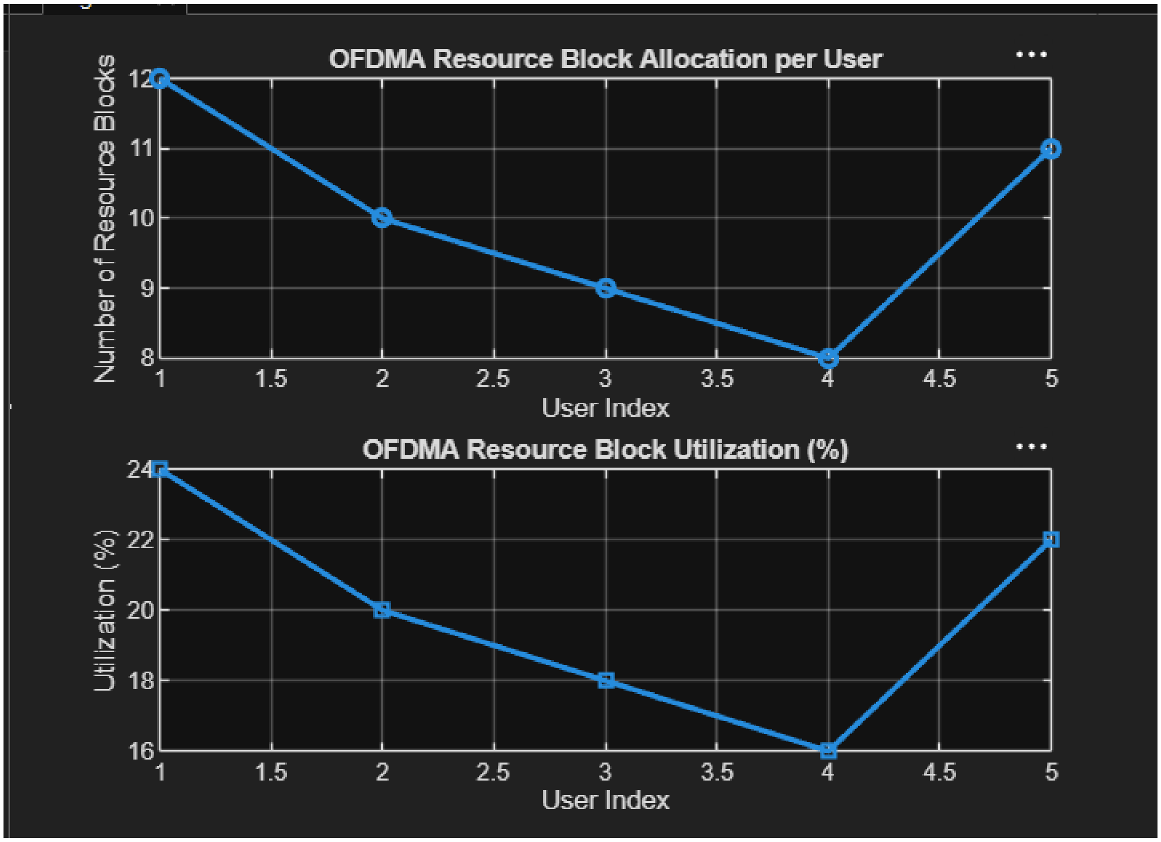
exp4_2_OFDMA_Resource_Block_Utilization.m

MATLAB Code

```
clc;

clear;

close all;


% Total Resource Blocks

total_RB = 50;


% Users

users = 1:5;


% Allocated RBs

alloc = [12 10 9 8 11];


% Utilization

util = (alloc/total_RB)*100;


disp('Resource Block Utilization (%) per User:')

disp(util')


figure
```

```
% Graph 1
subplot(2,1,1)
plot(users,alloc,'-o','LineWidth',2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on

% Graph 2
subplot(2,1,2)
plot(users,util,'-s','LineWidth',2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on
```

Output



- Resource Block Utilization (%) for each user.
- Graph of Resource Block Allocation.
- Graph of Resource Block Utilization.

Result

The resource utilization percentage was successfully calculated for each user. The results indicate efficient utilization of available spectrum resources in the OFDMA system.

Objective 3: User Throughput Analysis

Aim

To calculate the **User Throughput (Mbps)** based on the allocated Resource Blocks and modulation efficiency in an OFDMA system.

MATLAB File Name

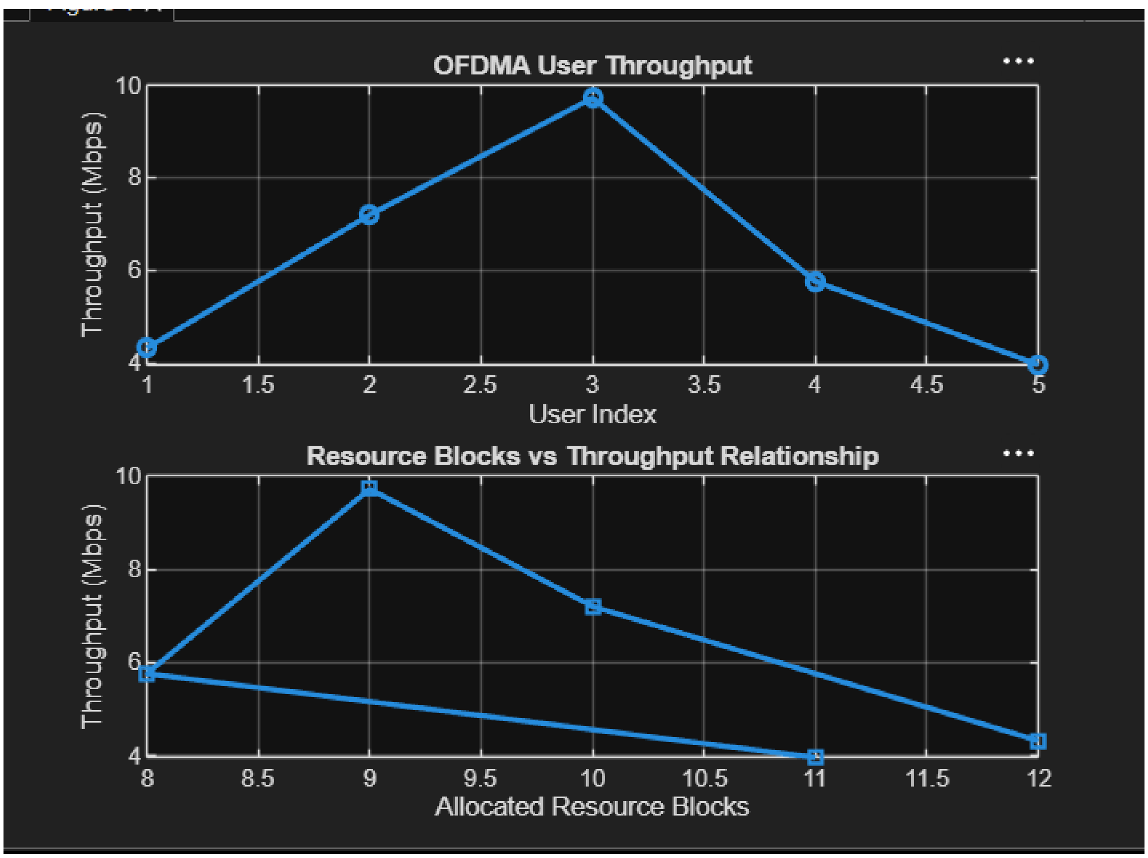
exp4_3_OFDMA_User_Throughput.m

MATLAB Code

```
clc;
clear;
close all;

% Bandwidth per Resource Block
rb_bw = 180e3;

% Users
users = 1:5;

% Allocated Resource Blocks
alloc_RB = [12 10 9 8 11];

% Modulation Efficiency
mod_eff = [2 4 6 4 2];

% Throughput Calculation
throughput = (alloc_RB .* rb_bw .* mod_eff)/1e6;

disp('User Throughput (Mbps):')
disp(throughput')

figure

% Graph 1
subplot(2,1,1)
plot(users,throughput,'-o','LineWidth',2)
```

```
title('OFDMA User Throughput')
xlabel('User Index')
ylabel('Throughput (Mbps)')
grid on

% Graph 2
subplot(2,1,2)
plot(alloc_RB,throughput,'-s','LineWidth',2)
title('Resource Blocks vs Throughput Relationship')
xlabel('Allocated Resource Blocks')
ylabel('Throughput (Mbps)')
grid on
```

Output



- User Throughput (Mbps).
- Graph of Throughput per User.
- Graph showing Resource Blocks vs Throughput.

Result

The user throughput was successfully calculated using the allocated resource blocks and modulation efficiency. The results show that users with more allocated resource blocks and higher modulation efficiency achieve greater throughput, demonstrating improved OFDMA system performance.